Zachary Porat

Department of Mathematics and Statistics
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U.S. Citizen
Last Updated November 1, 2026

Education

Wesleyan University — Middletown, CT
Ph.D. in Mathematics — 2020 – 2026

- Thesis: "On the Cuspidal Cohomology of Congruence Subgroups of $\mathrm{SL}(3,\mathbb{Z})$" [Link]
- Advisor: David Pollack

Union College — Schenectady, NY
B.S. in Mathematics with Honors, summa cum laude — 2016 – 2020

- Honors Thesis: "Classification of Torsion Subgroups for Mordell Curves" [Link]
- Physics Minor, Phi Beta Kappa, Scholars Program

Employment

Bucknell University — Lewisburg, PA
Visiting Assistant Professor — 2026 – Present

Research Interests

Broad: number theory, arithmetic geometry

Specific: automorphic forms, cohomology of arithmetic groups, arithmetic statistics, algebraic curves, computational aspects of number theory and arithmetic geometry

Publications and Preprints

[6] **On the Cuspidal Cohomology of Iwahori Congruence Subgroups of** $\mathrm{SL}(3,\mathbb{Z})$, submitted (2026). [Code]

arXiv:2609.01776

[5] **Distinguishing Elliptic Curves Modulo** p **and Identifying Images of Product Representations** (with Jessica Bennett, Jeffrey Hatley, Sasha Kononova, Vincent Macri, and Naina Praveen), submitted (2026). [Code]

arXiv:2608.10880

[4] **Heuristics for (Ir)reducibility of p-Rank Strata of the Moduli Space of Hyperelliptic Curves** (with Thomas Bouchet, Erik Davis, Steven R. Groen, and Benjamin York), *Experimental Mathematics* (2026). [Code]

DOI: 10.1080/10586458.2026.2671126

[3] **Computations Directly on the Cuspidal Cohomology of Congruence Subgroups of** $\mathrm{SL}(3,\mathbb{Z})$, *Mathematics of Computation* (2025). [Code]

DOI: 10.1090/mcom/4155

[2] **Family Sizes for Complete Multipartite Graphs** (with Danielle Gregg, Thomas Mattman, and George Todd), *Involve, a Journal of Mathematics*, Vol. 15, No. 4 (2022), 669-686.

MR4536581, DOI: 10.2140/involve.2022.15.669

[1] **PIXE Analysis of Synthetic Turf** (with Sajju Chalise, Morgan Clark, Skye Conlan, Scott LaBrake, and Michael Vineyard), *Environment and Ecology Research*, Vol. 6, No. 1 (2018), 60-65.

DOI: 10.13189/eer.2018.060105

Teaching Experience

Bucknell University

Instructor of Record Fall 2026 – Present

- Math 245 - Linear Algebra (Fall 2026)
- Math 201 - Calculus I (Fall 2026 $\times 2$)

Undergraduate Research Mentor Fall 2026 – Present

- Huaning Yuan - $\mathrm{GL}(3)$ Automorphic Forms and the LMFDB (Fall 2026)

Wesleyan University

Instructor of Record Fall 2022 – Spring 2026

- Math 500 - Graduate Pedagogy (Fall 2025)
- Math 121 - Calculus I (Fall 2024)
- Math 132 - Elementary Statistics (Fall 2022)

Graduate Teaching Assistant Fall 2021 – Spring 2026

- Math 221 - Vectors and Matrices (Spring 2026)
- Math 261 - Introduction to Abstract Algebra (Fall 2023, Fall 2025)
- Math 225 - Introduction to Real Analysis (Spring 2022, Spring 2023, Spring 2025)
- Math 262 - Advanced Topics in Abstract Algebra (Spring 2024)
- Math 228 - Discrete Mathematics (Fall 2021)

Directed Reading Program Mentor Spring 2023 – Spring 2025

- William Allen - Elliptic Curve Cryptography: Data Encryption through Math (Spring 2025)

- William Dollhopf - Introduction to Cryptography and the ElGamal Cryptosystem (Fall 2023)
- Emily Bennett - Understanding PKC: The Diffie-Hellman Key Exchange (Spring 2023)
- Stefan Hesseling - RSA and Integer Factorization (Spring 2023)

Math Workshop Tutor Fall 2020 – Spring 2021

- Covers the entire selection of undergraduate math courses at Wesleyan

Union College
Course Assistant Fall 2017 – Winter 2020

- MTH 340 - Linear Algebra (Winter 2020)
- MTH 110 - Calculus I (Fall 2017, Fall 2018, Fall 2019)
- MTH 112 - Calculus II (Winter 2018, Winter 2019)
- MTH 115 - Differential Vector Calculus and Matrix Theory (Spring 2018)

Calculus Help Center Tutor Fall 2019 – Winter 2020

- Covers Calculus I, Calculus II, and Multivariable Calculus with Linear Algebra

OUTREACH

Project Leader (with Jeffrey Hatley) June 22 – July 3, 2026
Rethinking Number Theory [Link] [See Paper #5] Virtual

Project Assistant: Iwasawa Theory and Applications May 5-9, 2025
Arithmetic Geometry at UNT [Link] Denton, TX

Mini-Course Lecturer: Mathematical Cryptography [Notes] March 29-30, 2025
Student Workshop in Mathematics [Link] Middletown, CT

INVITED TALKS

AMS Spring Eastern Sectional April 11, 2027
Special Session: Number Theory at PUIs Syracuse, NY

Joint Mathematics Meetings January 13, 2027
Special Session: Explicit Methods in Number Theory Chicago, IL

Lafayette College Math Colloquium October 29, 2026
Lafayette College Easton, PA

Bucknell University Math Colloquium April 3, 2026
Bucknell University Lewisburg, PA

CSUB Math Seminar March 27, 2026
California State University, Bakersfield Bakersfield, CA

UIC Number Theory Seminar March 13, 2026
University of Illinois Chicago Chicago, IL

Five College Number Theory Seminar February 10, 2026
Amherst College Amherst, MA

Wesleyan Algebra and Number Theory Seminar February 6, 2026
Wesleyan University Middletown, CT

UConn Algebra Seminar December 3, 2025
University of Connecticut Storrs, CT

AMS Fall Eastern Sectional Meeting October 25, 2025
Special Session: Number Theory Meets Algebraic Geometry Virtual

AMS Fall Southeastern Sectional Meeting October 4, 2025
Special Session: Modular Forms in Combinatorics and Number Theory New Orleans, LA

AWM Research Symposium May 16, 2025
Special Session: Curves and Abelian Varieties in Characteristic p Madison, WI

AMS New England Graduate Student Conference April 19, 2025
Brown University Providence, RI

Manhattan College Math Seminar November 15, 2023
Manhattan College Bronx, NY

Union College Math Seminar January 12, 2023
Union College Schenectady, NY

Contributed Talks

33èmes Journées Arithmétiques July 4, 2025
Université du Luxembourg Belval Esch-sur-Alzette, LUX

37th Automorphic Forms Workshop May 4, 2025
University of North Texas Denton, TX

Conférence de Théorie des Nombres Québec-Maine October 6, 2024
Université Laval Québec City, CAN

36th Automorphic Forms Workshop May 24, 2024
Oklahoma State University Stillwater, OK

Graduate Student Seminar March 9, 2023
Wesleyan University Middletown, CT

Graduate Student Seminar October 14, 2021
Wesleyan University Middletown, CT

Select Workshops and Conferences

Rethinking Number Theory[†] — June 22 – July 3, 2026
Distinguishing Elliptic Curves and Understanding Product Representations — Virtual

Arizona Winter School — March 7-11, 2026
Computational Aspects of Arithmetic Geometry and Cryptography — Tucson, AZ

LMFDB, Computation, and Number Theory (LuCaNT) 2025 — July 7-11, 2025
Institute for Computational and Experimental Research in Mathematics — Providence, RI

Arithmetic Geometry at UNT[‡] — May 5-9, 2025
University of North Texas — Denton, TX

Algorithmic Number Theory Symposium (ANTS) XVI — July 15-19, 2024
Massachusetts Institute of Technology — Cambridge, MA

Arizona Winter School — March 2-6, 2024
Abelian Varieties — Tucson, AZ

Professional Service

Reviewer for zbMath Open — Spring 2025 – Present
zbMath Open — Virtual

Workshop Organizer (with Lydia Ahlstrom and Arianna Zikos) — March 2025
Student Workshop in Mathematics [Link] — Middletown, CT

Departmental Service

DRP Organizer (with Arianna Zikos) — Spring 2023 – Spring 2025
Wesleyan Directed Reading Program [Link] — Middletown, CT

President — Fall 2023 – Spring 2025
Wesleyan AMS Graduate Student Chapter — Middletown, CT

Seminar Organizer (with Justin Bryant) — Spring 2022 – Fall 2023
Wesleyan Graduate Student Seminar — Middletown, CT

Vice President — Fall 2021 – Spring 2023
Wesleyan AMS Graduate Student Chapter — Middletown, CT

Honors and Awards

Martin Terry Resch Prize [Link] — May 2020
Union College, Department of Mathematics — Schenectady, NY

[†]Denotes that I served as a project leader at this workshop. See Outreach for additional details. See also Paper #5.
[‡]Denotes that I served as a project assistant at this workshop. See Outreach for additional details.

Phi Beta Kappa Membership May 2020
Union College Chapter Schenectady, NY

Pi Mu Epsilon Membership April 2019
Union College Chapter Schenectady, NY

Grants

AMS Sectional Travel Grant October 2025
AMS Fall Southeastern Sectional Meeting New Orleans, LA

AWM Research Symposium Travel Grant May 2025
AWM Research Symposium Madison, WI

Skills

Computer: Magma, SageMath, Python, Linux, LaTeX

Language: English (native), French (proficient)